

Assignment 3- Interactive prototype creation

Supermarket Inventory Management
System

Prepared By:
Ashwin.R

1. Introduction

For this assignment, I built an interactive prototype of an Inventory Management System in Figma. The goal was to simulate what it actually feels like to use a real stock management tool — tracking inventory, managing products, updating stock info, pulling reports, and moving between different sections of the app without it feeling clunky or disconnected.

Rather than just putting together static screens, I wanted the prototype to feel like something you could actually hand to a user and have them figure out on their own.

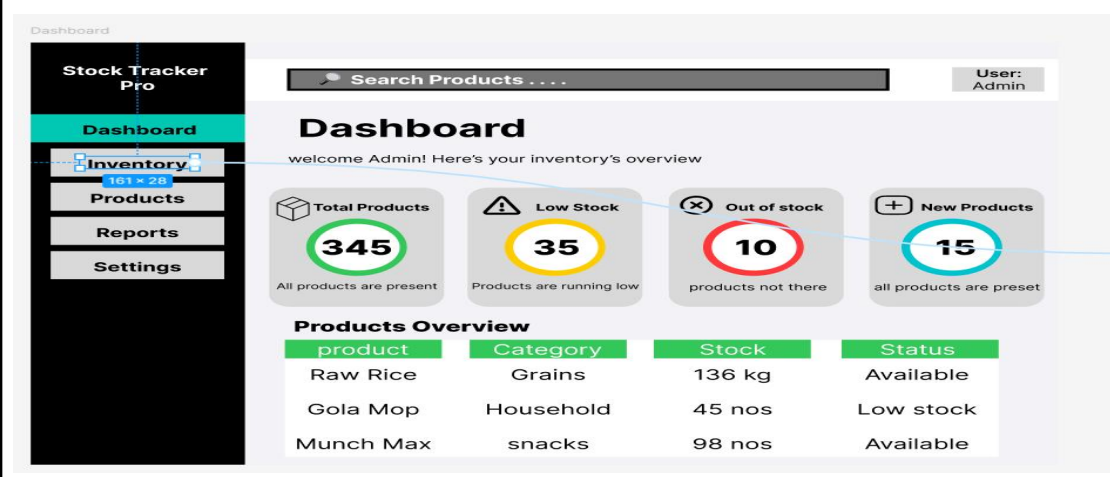
2. What the prototype does

The prototype walks through a full inventory management workflow — from checking your dashboard stats to adding a new product and viewing reports. Here's what's actually functional in it:

- Dashboard overview with live stock stats
- Inventory monitoring and product search
- Viewing and editing product details
- Adding new products to the system
- Accessing reports and analytics
- Confirmation flow after adding a product

3. The screens

Dashboard — The first thing you see when you open the app. It gives you a quick snapshot of total products, low stock items, out-of-stock counts, and a product overview table so you know where things stand at a glance.



Inventory Management — This is where the bulk of the work happens. You can search for products, filter by category, and see the full inventory table with stock status info for each item.

The screenshot shows the 'Inventory Management' screen. On the left is a sidebar with a 'Stock Tracker Pro' header and navigation links: 'Dashboard', 'Inventory' (highlighted), 'Products', 'Reports', and 'Settings'. A '2' badge is next to 'Inventory'. At the top right, there's a 'Search Products' bar and a 'User: Admin' indicator. The main area is titled 'Inventory Management' and contains a 'search Inventory' bar, a category dropdown set to 'All category', and an 'Apply filter' button. Below this is a table with the following data:

product	Category	Stock	Status
Raw Rice	Grains	96 nos	Available
Gola Mop	Household	45 nos	Low stock
Munch Max	Snacks	95 nos	Available
coca gola	Juice	0 nos	Out of stock
Gio Oi	Grocery	75 nos	Available

Product Details — Clicking on any product pulls up this screen, which shows everything about it — image, name, category, current stock, minimum stock threshold, and supplier info. You can also jump into editing from here.

The screenshot shows the 'Product description' screen. The sidebar is identical to the previous screen, but 'Products' is now highlighted. The main area is titled 'Product description' and features a 'Search Products' bar and a 'User: Admin' indicator. On the left is a large image of olive oil bottles. To the right of the image, the following details are listed:

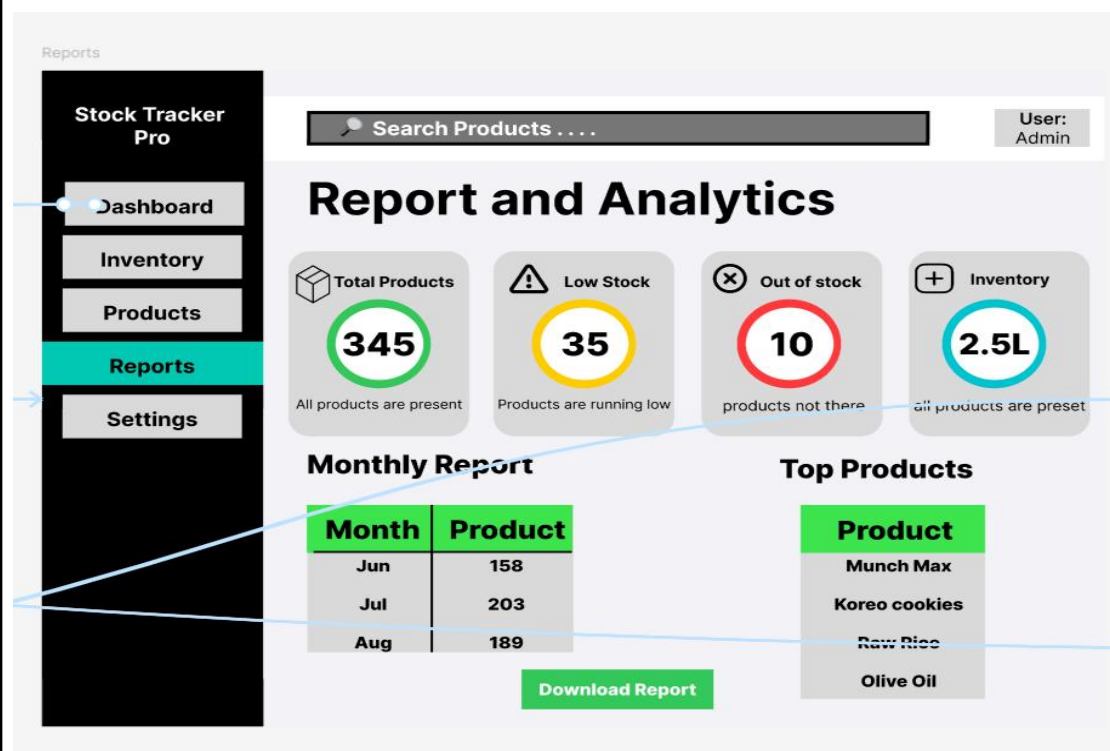
- Product Name: Olive oil
- Category: Grocery
- Current Stock: 96 nos
- Minimum Stock: 20 nos
- Supplier: Gio Foods
- Status: Available (indicated by a green dot)

At the bottom right, there is a green 'Edit Product' button. At the bottom left, there is a 'Back' button, and at the bottom right, there is a 'Next >' button.

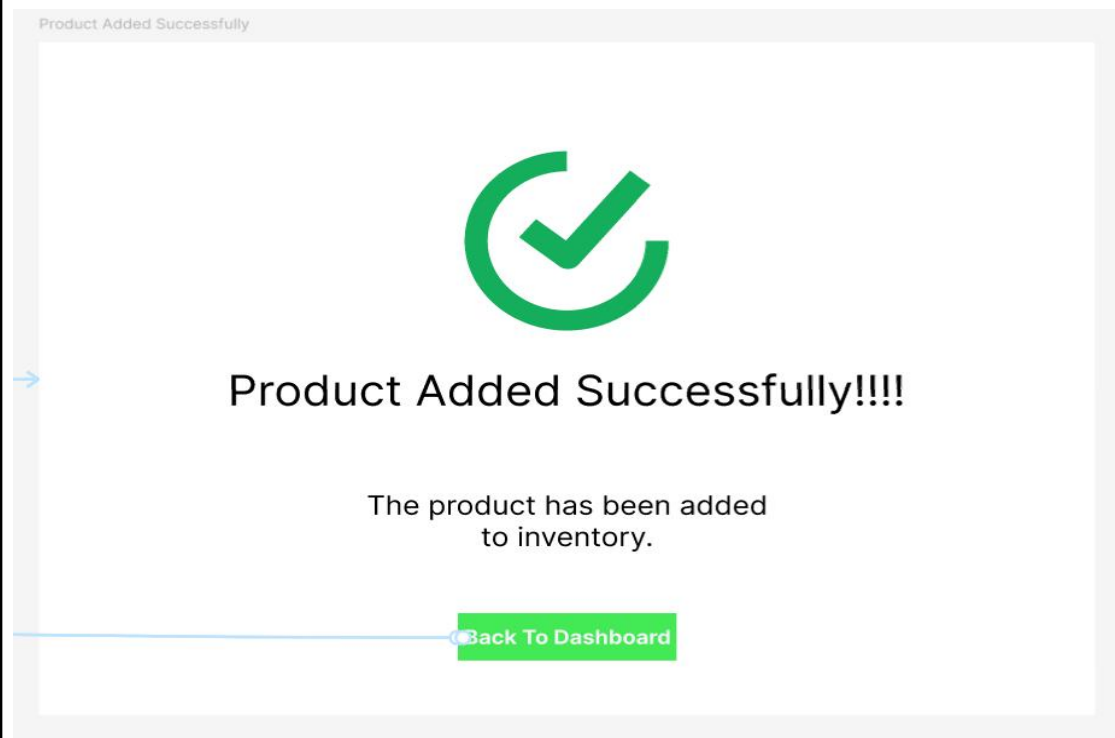
Add Product — A form screen where you can bring a new product into the system. Fields include product name, category, quantity, minimum stock, supplier details, and an image upload option.

The screenshot shows the 'Add new product' form in the 'Stock Tracker Pro' application. The left sidebar contains navigation links: Dashboard, Inventory, Products (highlighted), Reports, and Settings. The main content area has a search bar at the top right with the text 'Search Products ...' and a user profile 'User: Admin'. The form itself is titled 'Add new product' and includes the following fields: 'Product name' (text input), 'Category' (dropdown menu), 'Quantity' (text input), 'Minimum Stock' (text input), 'Supplier name' (text input), 'Unit' (dropdown menu), and an 'Upload Photo of the Product' section with an 'Upload File' button. A green 'Add product' button is at the bottom left of the form.

Reports & Analytics — A dedicated section for viewing inventory summaries, monthly data, and top-performing products. There's also a download option if you need to export the report.



Product Added Successfully — A simple confirmation screen that appears after a new product is added, with an option to return to the dashboard.



4. How the navigation works

The flow follows the natural way someone would actually use an inventory system day-to-day:

Dashboard → Inventory Management → Product Details → Add Product → Reports & Analytics → Confirmation → Dashboard

It loops back to the dashboard so users can keep working without hitting any dead ends.

5. Interactions Built-in

- The **sidebar** lets you jump between major sections of the app at any point
- Clicking a product in the table **opens its detail view**
- From details, you can **edit stock information**
- The **Add Product form** lets you create a new entry from scratch

- The **Reports section** is accessible directly and shows real data layouts
- After adding a product, you get a **confirmation screen** so nothing feels like it disappeared into a void

6. Animations and Transitions

All page transitions use Figma's Smart Animate at 300ms. It's a small detail, but it makes moving between screens feel intentional rather than jarring — especially important for a tool people would be using repeatedly throughout a workday.

7. User Testing

I ran a few users through the prototype with five tasks: check the inventory stats, search for a product, open its details, add a new product, and access the reports section.

Overall it went well:

- All users finished every task without needing help
- Navigation through the sidebar felt natural and easy to pick up
- Product info was laid out clearly and wasn't hard to scan
- The add product workflow was smooth end-to-end
- Reports were straightforward to find and read

8. Conclusion

This was a useful project for understanding how prototyping works beyond just making things look nice. Thinking through the actual user flow, watching people test it, and then going back to fix what didn't work gave me a much clearer picture of user-centered design in practice. The final prototype covers the core inventory management workflow and came out of testing in pretty solid shape.